In this potion of the chapter, we will discussthe most basic and essential commands which are frequently used.

ping - Send an ICMP ECHO_REQUEST to network hosts

The ping command is one of the most used utilities for troubleshooting, testing, and diagnosing network connectivity issues. Ping works by sending one or more ICMP (Internet Control Message Protocol) Echo Request packages to a specified destination IP on the network and waits for a reply. To understand how ping command works, let us ping *www.google.com*

```
$ ping google.com
```

Following is the output

```
PING google.com (172.217.22.206) 56(84) bytes of data.
64 bytes from muc11s01-in-f14.1e100.net (172.217.22.206): icmp_seq=1 ttl=53 time=40.2 ms
64 bytes from muc11s01-in-f14.1e100.net (172.217.22.206): icmp_seq=2 ttl=53 time=41.8 ms
64 bytes from muc11s01-in-f14.1e100.net (172.217.22.206): icmp_seq=3 ttl=53 time=47.4 ms
64 bytes from muc11s01-in-f14.1e100.net (172.217.22.206): icmp_seq=4 ttl=53 time=41.4 ms
^C --- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 7ms
rtt min/avg/max/mdev = 40.163/42.700/47.408/2.790 ms
```

What we understand from the output-

- The number of data bytes is 56 by default, which translates into 64 ICMP data bytes - 64 bytes and the IP address of the destination - from muc11s01-in-f14.1e100.net (172.217.22.206)
- The ICMP sequence number for each packet. icmp_seq=1
- The Time to Live. - ttl=53
- The ping time, measured in milliseconds is the round trip time for the packet to reach the host, and its response to return to the sender. - time=41.4 ms

By default, the interval between sending a new packet is one second.

The ping command will continue to send ICMP packages to the Destination IP address until it receives an interrupt. To stop the command, just hit the Ctrl+C key combination. Once the command stops, it displays a statistic including the percentage of packet loss. The packet loss means the data was dropped somewhere in the network, indicating an issue within the network. If there is a packet loss, you can use the traceroute command to identify where